Honest Review: Novelty, Standing and Publication Path

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1. Short answer

Yes: the work has three genuine novelties and two supporting contributions. They are a new finding, a new attack analysis and a simple composed defence, not a new detector.

Your question	Honest answer
Genuine novelties? How many?	Yes. Three core ones: the label-channel finding (N1), verified-label learning with uncertainty-directed audits (N2), and the framing attack with its consensus fix (N3). Two supporting ones: a corrected re-evaluation of MSAAD (S1) and negative results on existing defences (S2).
Have we surpassed the original paper?	As an evaluation, and on threats it never tested: yes. As a detector: no. We reuse MSAAD's detector, clean accuracy is at parity, and MSAAD still leads on speed reporting and detector baselines.
Are our results better?	Only where the label channel is attacked. Under full label poisoning, evasion stays at or below 1.2%, where MSAAD's learning rule lets 99.7–99.96% of attacks through. Some settings cost us more false alarms.
Where do we stand against recent research?	Among online IoT DDoS detection papers, ours is the only one we found that tests late and framed labels as well as poisoned ones and defends with verified labels, and it is more rigorous than the closest works. The underlying ideas are already known in wider ML security.
Publishable as a successor?	Yes, framed as a follow-up that re-evaluates and hardens MSAAD, not as a better MSAAD.

Your question	Honest answer
At which platform?	Now, after small fixes: IEEE Access, Computers & Security, IEEE CNS, or an IEEE ICC or GLOBECOM security symposium. With 3–6 more months: ACSAC, RAID or AsiaCCS. The top four security conferences: unlikely.
Good reviews as it is?	At a mid-tier venue, borderline to weak accept, most likely with major revisions. At a top venue, a reject.

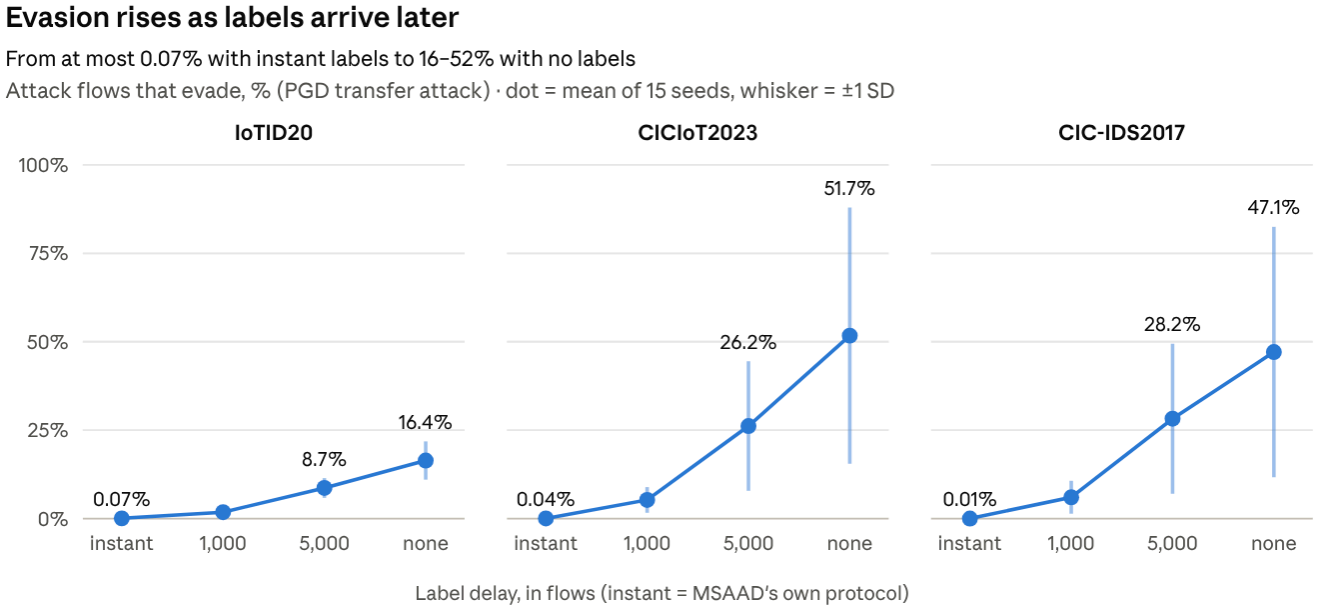
2. The three genuine novelties

Each one is new for online IoT DDoS detection, but none is new as a general machine-learning idea. Tell the professor both halves, so a reviewer cannot say the second one first. Triples read IoTID20 / CICIoT2023 / CIC-IDS2017.

N1. MSAAD's robustness comes from its label channel, not its model

What it is. MSAAD's Algorithm 1 hands the detector each flow's true label right after it predicts. We showed that this instant, correct label stream is what stops evasion.

Evidence. With instant labels, no attack gets more than 0.12% of its flows through. Without labels, valid-traffic PGD evades 17–49% of the time (Holm-adjusted $p \leq 0.0014$), and a 5,000-flow delay already lets 9–28% through.



Our paper, Table III (raw PGD rows) · 15 seeds per point

The chart uses raw PGD, the attack the delay runs used; repaired PGD gives 17–49% with no labels, and the paper finds no significant difference between the two. The wide whiskers on CICIoT2023 and CIC-IDS2017 mean transfer depends on each seed's surrogate model: some seeds evade almost fully, others barely.

Already known. Poisoning of online learners and delayed labels in data streams are studied problems (Zhang et al. 2020; Margiotta et al. 2023; Grzenda et al. 2020; Ceschin et al. 2024).

New here. A measured, significance-tested demonstration that a published IoT DDoS defence loses its robustness once its labels are late or missing.

N2. Verified-label learning with uncertainty-directed audits

What it is. Four rules decide which labels the detector may learn, with 10% of flows audited:

1. Never learn an unverified benign label, because the attacker's goal is to get its own flows labelled benign.
2. Learn each audited benign flow 10 times, to make up for the benign labels skipped (IW-strict).
3. Learn an unverified DDoS label only when the detector agrees with it (IW-consensus).
4. Spend the audits on the flows the ensemble is least sure about.

Evidence. Under full label poisoning, evasion stays at or below 1.2%, against 99.7–99.96% for MSAAD's learning rule. Uncertainty-directed audits cut false alarms under attack 8–13× against random audits, to below MSAAD's own rate, while using fewer audits. Section 6 has the full numbers.

Already known. Each ingredient: trusted-label correction (Hendrycks et al. 2018), importance-weighted active learning (Beygelzimer et al. 2009), uncertainty sampling (Settles 2009) and agreement filters.

New here. One-sided trust derived from the attacker's goal, the combination inside an online network detector, and an ablation showing what each part buys.

Weak spot. On clean IoTID20 traffic, uncertainty audits raise false alarms at 99.9% detection from 3.46 to 10.39%, and we cannot yet explain why.

N3. Framing: the counter-attack that verification cannot stop

What it is. Once a defence trusts DDoS labels, the attacker can send benign-looking flows from its bots. Those flows are truly labelled DDoS, so audits confirm them, and the detector learns to flag genuine benign traffic.

Evidence. At 2% framing, the strict policy flags 61.3 / 92.3 / 22.6% of genuine benign flows, against 11.1 / 25.5 / 2.5% for MSAAD's rule. IW-consensus with uncertainty audits is never significantly worse than MSAAD's rule, during or after the campaign.

Already known. Attacks that trick a learner into blocking legitimate traffic: the Allergy attack (Chung and Mok 2006) and causative availability attacks (Barreno et al.; Nelson et al. 2008).

New here. Framing as the natural answer to one-sided label trust, the reason verification cannot catch it (its labels are true), and a consensus check that contains it.

3. Two supporting contributions

Neither is a new idea, but both change what readers should believe about MSAAD, and reviewers value that kind of evidence.

S1. A corrected re-evaluation of MSAAD

- **Valid attacks.** We perturb only the attacker's own flows and repair each one to valid traffic. MSAAD perturbs benign flows too; doing the same gives 65 / 78% false alarms (IoTID20 / CICIoT2023), which explains its reported precision collapse.
- **Security metrics.** We report evasion (ASR) and false alarms (FPR); MSAAD reports neither. Always answering "DDoS" already scores 87.6 / 96.6 / 75.4% accuracy on these streams.
- **Statistics.** 15 seeds, paired Wilcoxon tests with Holm correction, bootstrap intervals and effect sizes on every comparison.
- **Dataset audit.** IoTID20's public file is shuffled (only 50% of consecutive rows are in time order), so MSAAD's three "concept drifts" on it cannot be real. We added a time-ordered CIC-IDS2017 stream.
- **Reproduction.** Clean accuracy matches: 99.33 vs 99.50% on IoTID20 and 99.76 vs 99.28% on CICIoT2023. MSAAD's CICIoT2023 collapse reproduces only without labels. Its IoTID20 collapse (38.92–67.12%) did not reproduce: even perturbing every flow without labels left 78.0% accuracy.
- **Found but not yet in the paper.** MSAAD's "recovered accuracy" counts only the flows RADP lets through: it reads 99.8% on clean CICIoT2023 while 22% of benign traffic is lost. Its GAN samples are fully distinguishable from real adversarial flows (two-sample test AUC 1.000). Its CICIoT2023 F1 of 97.98 does not match its own precision and recall, which give 97.69.

S2. Negative results on existing defences

- **RADP, MSAAD's GAN filter,** mostly catches impossible feature values rather than attacks. On IoTID20 it flags 91–100% of raw attacks but only 11–17% of repaired ones; on CICIoT2023 it deletes 26% of genuine benign flows. When it has not trained on C&W, it misses most repaired C&W flows (project record, not in the paper's table).
- **Adversarial training** stops transfer evasion on CICIoT2023 (48.52 → 0.03%) but more than doubles false alarms (6.41 → 14.13%). A label-only HopSkipJump attacker still evades 14.0 / 2.0% of the time (IoTID20 / CICIoT2023).
- **A graph neural network stage** is never significantly better than a parameter-matched MLP on per-host features. Spoofing 10 source identities raises evasion after the full chain from 12.47 to 18.89% on IoTID20.

4. What is not new

State these limits up front in any presentation, because an expert reviewer will check them first.

- **The detector.** We rebuilt MSAAD's layers 2 and 3 in River and propose no new classifier or architecture.
- **The threat ideas.** Poisoning online learners, label flipping and trusted-label defences are established ML-security topics.
- **The attacks.** FGSM, BIM, PGD, C&W and HopSkipJump are standard, and feasibility repair follows the problem-space work of Pierazzi et al. (2020) and Apruzzese et al. (2022).
- **The weighting.** Fixed copies of audited flows simplify importance-weighted active learning; they are not a new weighting scheme.
- **The data.** All three datasets are public lab captures, and we collected no new traffic.

5. Have we surpassed the original paper?

We surpass MSAAD as an evaluation and on every threat it never tested. We do not surpass it as a detector: it is the same detector, and MSAAD still leads on detector baselines and speed reporting.

Dimension	MSAAD (IEEE Access 2025)	Our paper	Ahead
Threat model	Evasion only; labels instant and correct	Evasion plus late, missing, poisoned and framed labels, with an audit budget	Ours
Attack realism	All flows perturbed, no feature constraints	Attacker's flows only, repaired to valid traffic; label-only HopSkipJump	Ours
Metrics	Accuracy, precision, recall, F1	Evasion (ASR), false alarms (FPR), ROC AUC, FPR at matched detection, audits used	Ours
Statistics	Confidence interval and p-value for clean accuracy only	15 seeds; paired tests with Holm correction, bootstrap intervals and effect sizes, including under attack	Ours
Datasets	IoTID20, CICIoT2023	The same two plus a time-ordered CIC-IDS2017	Ours
Detector design	New three-layer architecture	None; reuses MSAAD's detector	MSAAD
Detector baselines	Seven other online learners	Update policies, RADP, adversarial training, detector chain	MSAAD
Speed	Learning time, latency, throughput (719 flows/s on CICIoT2023)	Not reported	MSAAD
Clean accuracy	99.50 / 99.28%	99.33 / 99.76% (our rebuild)	Tie
Evasion with instant labels	About 99% accuracy after RADP	At most 0.12% evasion, even without RADP	Tie
Late or missing labels	Not tested	Measured, not fixed	Ours (measured only)
Poisoned labels	Not tested	Evasion at or below 1.2%	Ours

Dimension	MSAAD (IEEE Access 2025)	Our paper	Ahead
Framing	Not tested	Identified and contained, at a false-alarm cost	Ours
Code	Not released	Available on request, not yet public	Neither

6. Are our results better?

In MSAAD's own setting the results are equal, because the detector is the same. Under attacks on the label channel, ours are far better on evasion and mixed on false alarms. Values are % for IoTID20 / CICIoT2023 / CIC-IDS2017, means over 15 seeds, from Tables II and IV–VI of our paper.

Setting	Measure	MSAAD's learning rule	Our policy	Verdict
Clean traffic, instant labels	Accuracy	99.54 / 99.71 / 99.93	Same detector	Tie
Full poisoning plus evasion	Evasion	99.74 / 99.96 / 99.95	1.15 / 0.43 / 0.13 (IW-strict, random audits)	Ours
Evasion with true labels	False alarms	2.17 / 3.52 / 0.11	0.45 / 0.64 / 0.02 (IW-strict, uncertainty audits)	Ours
Evasion with true labels	False alarms	2.17 / 3.52 / 0.11	3.52 / 6.95 / 0.25 (IW-strict, random audits)	MSAAD
Clean traffic, 99.9% detection	False alarms	2.90 / 7.55 / 0.00	10.39 / 5.81 / 0.01 (IW-strict, uncertainty audits)	Mixed; worse on IoTID20
Poisoning, 5% framing and evasion	Evasion	99.74 / 99.93 / 99.95	0.65 / 0.35 / 0.15 (IW-consensus, uncertainty audits)	Ours
Poisoning, 5% framing and evasion	False alarms	8.75 / 29.83 / 3.51	5.23 / 13.75 / 1.95	Ours

Setting	Measure	MSAAD's learning rule	Our policy	Verdict
Poisoning and 5% framing, no evasion	Evasion	99.61 / 99.72 / 99.84	1.67 / 0.24 / 0.14	Ours
Poisoning and 5% framing, no evasion	False alarms	7.16 / 25.01 / 3.26	11.27 / 26.07 / 2.65	Mixed; worse on IoTID20
Late or missing labels	Evasion	See the N1 chart	Not addressed	Open for both

Two caveats belong in any comparison. Every number is against our rebuild of MSAAD, because its code is not public. And under the combined attack, false alarms still rise 2.4–22× above normal operation; that is the price of keeping evasion low.

7. Where we stand against recent research

Within online IoT DDoS detection, ours is the only work we found that tests late and framed labels as well as poisoned ones and defends with verified labels, and it is the most rigorous of the close comparisons. In wider ML security these ideas are known, so the contribution is applied and empirical.

Work	What it does	Label channel	Relation to ours
Beshah et al. (MSAAD), IEEE Access 2025	Online ensemble with a GAN filter; IoTID20 and CICIoT2023; unconstrained evasion	Instant and trusted	The system we re-evaluate and harden
Nwokoye, El-Khatib and Yang, arXiv, Oct 2026	Online AutoML in River on IoTID20; random label flips and feature noise at rates up to 1; adversarial training, best F1 0.904 at full flipping	Labels used as delivered; no delay, verification or framing	Closest competitor; no seed count or significance tests; not numerically comparable (F1, random flips)

Work	What it does	Label channel	Relation to ours
Nwokoye, Zaheer, El-Khatib and Yang, arXiv 2026	The same pipeline under black-box evasion with limited feature checks; adversarial training	Instant and trusted	Evasion-only counterpart; no seed count or significance tests
Raza et al., Scientific Reports, Aug 2026	Liquid neural network; CICIoT2023, SNT-SDN2025 and IoT23; MIM evasion; reports FPR	Not studied; batch training	Cites MSAAD in related work only, with no experimental comparison
Severi et al., ACSAC 2023	Clean-label backdoors in batch-trained network flow classifiers	Training data poisoned	Shows flow-poisoning work reaches strong security venues
Margiotta et al., NeurIPS 2023	Theory and experiments on attacks against online learners	Stream poisoned	Theoretical backing for N1; must cite
Kloft and Laskov, JMLR 2012	Poisoning of online centroid anomaly detection	Stream poisoned	Early security analysis of online learning; must cite
Korycki and Krawczyk, Machine Learning 2023	Poisoning that fools drift detectors in data streams, and a robust drift detector	Stream poisoned	Closest stream-mining defence; must cite
Lin et al., GLOBECOM 2021	Active learning under malicious mislabelling, on image data	Labels poisoned	Same idea family as N2 in another domain; already cited
Zhi et al., arXiv 2025	Clean-label backdoors through active-learning sample selection	Selection exploited	The adaptive attacker our audits must face next; already cited

We lead the IoT DDoS papers on threat model and rigour, and we are the only one with verified labels. We trail top-venue ML-security work on method novelty and attacker modelling. Label-poisoning work outside streams, such as the FLIP attack (Jha et al. 2023) and the FLORAL defence (Bal et al. 2025), should also be cited.

Both 2026 preprints come from one Ontario Tech group working in exactly this niche. Cite them, expect them as possible reviewers, and treat them as competitors on timing.

8. Is it publishable as a successor?

Yes, as a follow-up that re-evaluates and hardens MSAAD, not as a better MSAAD, because the detector is unchanged. The current title already frames it that way, and it should stay that way.

Tier	Example venues	Ready?	What it needs
Mid-tier journal	IEEE Access (MSAAD's own venue), Computers & Security, Journal of Information Security and Applications	After small fixes	Missing related work, the claim fixes in Section 9, and the extra findings from S1
Mid-tier conference	IEEE CNS; IEEE ICC or GLOBECOM security symposia	After small fixes	The same, plus a cut to about 6 pages for ICC or GLOBECOM
Workshop	ACM AISec, held with CCS	After small fixes	A tighter focus on N1 and N3
Strong security conference or journal	ACSAC, RAID, AsiaCCS, ESORICS; IEEE TNSM, IEEE Internet of Things Journal	In 3–6 months	The upgrades below
Top four conferences	IEEE S&P, ACM CCS, USENIX Security, NDSS	Unlikely	A new insight backed by real-world label data across many detectors

How far it can go with more time. With 3–6 months of focused work, it has a fair shot at ACSAC or RAID, which fit the topic best. These upgrades matter most, in order:

1. Test 2–3 more online detectors, for example PWPAE, a plain ARF and a label-free detector such as Kitsune, so the finding covers online detectors in general.
2. Build a real labeller: replay low-rate floods through Suricata or Snort, and measure how often attack flows get benign labels and how late labels arrive.
3. Add an adaptive attacker that targets the audit rule, for example by keeping its flows where the ensemble is confident.
4. Replace fixed copies with randomized audits and 1/p weights (true IWAL), and explain or fix the clean-IoTID20 false-alarm rise.

5. Report audit cost (analyst time per audit), per-flow latency and throughput, as MSAAD does.
6. Vary the data sample with several windows per dataset, and add a newer IoT dataset with more benign traffic.
7. Release the code and configurations; artifact badges help at ACSAC and RAID.

Practical points

- Ask your professor to lead the venue choice and co-author; a course project needs their sign-off before submission.
- MSAAD's authors may review it, especially at IEEE Access, so keep the tone factual and credit what MSAAD does well.
- Disclose AI assistance as the venue requires; IEEE asks for it in the acknowledgments.
- Post a preprint once your professor agrees, because another group is publishing in this niche.

9. Would it get good reviews as it is?

At a mid-tier venue, borderline to weak accept, most likely with major revisions. At a top venue, a reject, mainly for limited method novelty and a single rebuilt system.

What reviewers will like

- A clear question with a surprising answer: the defence works only because its labels are perfect.
- Rigour above the norm for this literature: valid attacks, security metrics, 15 seeds and corrected tests.
- Honest limits and negative results, including where our own defence costs false alarms.
- A simple policy with a large effect: evasion falls from 99.7–99.96% to at most 1.2%.

What reviewers will criticise

Likely criticism	Severity	Fix	Effort
The defence combines known parts	High at top venues, low at mid-tier	Present it as an applied, empirical paper and lead with the ablation	Writing
One detector, rebuilt without the original code; the IoTID20 collapse did not reproduce	High	Ask the authors for their code; add 2–3 detectors	Weeks

Likely criticism	Severity	Fix	Effort
The attacker never adapts to the audit rule	High	Add an adaptive attacker	Weeks
Is full poisoning or a 10% audit budget realistic?	Medium to high	Measure label errors and delays with a real labeller; model audit cost	Weeks
2–26% false alarms under the combined attack; the clean-IoTID20 rise is unexplained	Medium	Randomized 1/p weights; investigate the IoTID20 case	Days to weeks
Lab data, about 330 benign CICIoT2023 test flows, seeds that do not vary the data	Medium	Several data windows; a newer dataset	Days
Missing related work	Medium, but easy	Cite the "must cite" works in Section 7, both Nwokoye preprints, Barreno et al. and Nelson et al.; soften "We are not aware of prior work"	Hours
No speed figures, while MSAAD reports them	Low to medium	Report per-flow latency and throughput	Days
Very dense, with many numbers per sentence	Low	Use journal length and move detail into tables	Writing

Quick fixes for the current paper

- The abstract and introduction say poisoning raises evasion to 99.7–99.95%, but Table IV shows 99.96% on CICIoT2023, so write 99.7–99.96%.
- Table III's delay rows (1,000 and 5,000 flows, 60 s and 300 s) use raw PGD, not repaired PGD, so label them that way. Your finding that raw and repaired attacks transfer equally well (Holm-adjusted $p \geq 0.46$) keeps the comparison valid.
- Three CHECK notes are now verified against MSAAD's text: the 38.92–67.12% and 13.08–32.38% ranges, "Resilient Adversarial Detection and Purification" for RADP, and "Accuracy Update Weighted Probability Averaging Ensemble" for AUWPAE. Only the Guard line, which you write, remains.

10. One-minute version for the professor

Five points to say in your own words, in this order:

1. MSAAD reports that its filter restores about 99% accuracy under attack, but its experiments hand the detector every flow's true label instantly.
2. We rebuilt it and showed that this label stream, not the model, stops evasion. Without labels 17–49% of valid attacks get through, and if the attacker gets its flows labelled benign, 99.7–99.96% do.
3. Our defence learns only what it can verify: no unverified benign labels, extra weight for audited benign flows, DDoS labels only on agreement, and audits aimed at uncertain flows. Evasion stays at or below 1.2%.
4. We also found the counter-attack, framing, which verification cannot stop, and a consensus rule that contains it at a false-alarm cost.
5. That makes three novelties (the finding, the defence and the framing analysis) plus a corrected re-evaluation and negative results. It is publishable now as a follow-up at a mid-tier venue; a top security venue would need more detectors, a real labeller and an adaptive attacker.

11. Sources

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- Zhi et al., "Selection-Based Vulnerabilities: Clean-Label Backdoor Attacks in Active Learning", arXiv 2508.05681, 2025
- Jha, Hayase and Oh, "Label Poisoning is All You Need", NeurIPS 2023
- Bal, Cevher and Muehlebach, FLORAL, arXiv 2502.17121, 2025
- Our numbers: the paper's Tables II–VII, its per-seed results, and the project record's findings F3, F8 and F13.